

Name: D. Sri Anjaneya
Register Number: 192425219

CO2 – AT2
MLA 0201 – Fundamentals of Machine Learning

	Annexure A		
	Scenario-Based		
Q1	Apply a linear regression model to predict apartment rent based on floor area, number of rooms, and location in a real-estate scenario.	5 Marks	6 min
Q2	Use a linear classification model to categorize emails as spam or non-spam using word frequency features in a mail service system.	5 Marks	6 min
Q3	Apply the Naïve Bayes algorithm to predict whether a patient has flu based on symptoms such as fever, cough, and headache.	5 Marks	6 min
Q4	Use discriminant functions to classify loan applicants into eligible and non-eligible groups based on income and credit score.	5 Marks	6 min
Q5	Apply a probabilistic generative model to classify handwritten digits in an optical character recognition system.	5 Marks	6 min
Q6	Implement a probabilistic discriminative model to predict customer churn from service usage data in a telecom scenario.	5 Marks	6 min
Q7	Apply Bayesian logistic regression to predict whether a customer will purchase a product after viewing online advertisements.	5 Marks	6 min
Q8	Use linear regression to forecast monthly sales for a retail company based on advertising expenditure and seasonal demand.	5 Marks	6 min
Q9	Apply Naïve Bayes classification in a news portal to categorize articles into sports, politics, and entertainment.	5 Marks	6 min
Q10	Use Bayesian logistic regression to predict student placement outcomes based on CGPA, coding skills, and aptitude scores.	5 Marks	6 min

1. Apartment Rent Prediction Using Linear Regression

Introduction

Linear Regression is a supervised machine learning algorithm used to predict continuous numerical values. In a real-estate scenario, it can estimate the rent of an apartment based on factors such as floor area, number of rooms, and location.

Input Features

- Floor Area (in square feet)
- Number of Rooms
- Location (encoded as numerical values)

Output

- Predicted Apartment Rent (₹)

Sample Dataset

Floor Area (sq.ft)	Rooms	Location	Rent (₹)
800	2	1	15,000
1000	3	2	22,000
1200	3	2	25,000
1500	4	3	35,000
900	2	1	17,000

(Location Encoding: 1 = Suburban, 2 = City, 3 = Premium Area)

Working

1. Collect apartment rental data.
2. Clean and preprocess the dataset.
3. Convert location into numerical values.
4. Train the Linear Regression model using the dataset.
5. The model learns the relationship between input features and rent.
6. For a new apartment, the model predicts the expected monthly rent.

Linear Regression Equation

$$\text{Rent} = b_0 + b_1(\text{Floor Area}) + b_2(\text{Rooms}) + b_3(\text{Location})$$

Where:

- b_0 = Intercept
- b_1, b_2, b_3 = Model coefficients

Example Prediction

Suppose a new apartment has:

- Floor Area = **1300 sq.ft**
- Rooms = **3**
- Location = **Premium Area (3)**

The trained model predicts the monthly rent as:

Predicted Rent = ₹29,500

Advantages

- Simple and easy to implement.
 - Provides accurate predictions for linear relationships.
 - Helps landlords and tenants estimate fair rental prices.
 - Supports real-estate pricing decisions.
-

2. Email Spam Detection Using Linear Classification

Introduction

A Linear Classification model is a supervised machine learning algorithm used to classify data into categories. In a mail service system, it is used to identify whether an email is **spam** or **non-spam (ham)** based on the frequency of words in the email.

Input Features

- Word frequency (e.g., "free", "offer", "win", "click")
- Number of links
- Number of special characters

Output

- Spam
- Non-Spam

Sample Dataset

Free	Offer	Click	Win	Email Type
5	3	2	1	Spam
0	0	0	0	Non-Spam
4	2	3	2	Spam
1	0	0	0	Non-Spam

Working

1. Collect labeled email data.
2. Extract word frequency features from each email.
3. Train the linear classification model using the dataset.
4. The model learns a decision boundary between spam and non-spam emails.
5. When a new email arrives, the model predicts whether it is spam or non-spam.

Example Prediction

Suppose a new email contains:

- Free = 4
- Offer = 2
- Click = 3
- Win = 1

The model predicts:

Email Type = Spam

Advantages

- Fast and efficient for email filtering.
- Easy to implement.
- Works well with text-based features.
- Reduces unwanted spam emails.

Conclusion

A Linear Classification model accurately classifies emails as spam or non-spam using word frequency features, improving email security and user experience.

3. Flu Prediction Using Naïve Bayes

Introduction

Naïve Bayes is a supervised machine learning algorithm based on Bayes' Theorem. It is commonly used for classification problems. In healthcare, it can predict whether a patient has the flu based on symptoms such as fever, cough, and headache.

Input Features

- Fever
- Cough
- Headache

Output

- Flu
- No Flu

Sample Dataset

Fever	Cough	Headache	Flu
Yes	Yes	Yes	Yes
Yes	No	Yes	Yes
No	Yes	No	No
No	No	No	No

Working

1. Collect patient symptom data.
2. Train the Naïve Bayes model using the dataset.
3. Calculate the probability of flu based on each symptom.
4. Compare the probabilities of **Flu** and **No Flu**.
5. Predict the class with the highest probability.

Example Prediction

Suppose a patient has:

- Fever = Yes
- Cough = Yes
- Headache = Yes

The model predicts:

Result = Flu

Advantages

- Simple and easy to implement.
- Works well with small datasets.
- Fast prediction.
- Effective for medical diagnosis.

Conclusion

Naïve Bayes efficiently predicts whether a patient has the flu by analyzing symptoms such as fever, cough, and headache, helping doctors make quicker diagnostic decisions.

4. Loan Eligibility Classification Using Discriminant Functions

Introduction

Discriminant Functions are classification techniques used to separate data into different groups. In a banking system, they help classify loan applicants as **eligible** or **non-eligible** based on factors such as income and credit score.

Input Features

- Annual Income
- Credit Score

Output

- Eligible
- Non-Eligible

Sample Dataset

Income (₹)	Credit Score	Loan Status
6,00,000	780	Eligible
3,50,000	620	Non-Eligible
8,00,000	810	Eligible
4,00,000	650	Non-Eligible

Working

1. Collect applicant data.
2. Use income and credit score as input features.
3. Train the discriminant function using the dataset.
4. The model finds a boundary that separates eligible and non-eligible applicants.
5. Predict the loan status for a new applicant.

Example Prediction

Suppose a new applicant has:

- Income = ₹7,00,000
- Credit Score = 790

The model predicts:

Loan Status = Eligible

Advantages

- Simple and effective for classification.
- Helps banks make quick loan decisions.
- Reduces manual effort.
- Improves decision accuracy.

Conclusion

Discriminant Functions classify loan applicants into eligible and non-eligible groups using income and credit score, enabling faster and more reliable loan approval decisions.

5. Handwritten Digit Classification Using a Probabilistic Generative Model

Introduction

A Probabilistic Generative Model is a machine learning model that learns the probability distribution of data and uses it for classification. In an Optical Character Recognition (OCR) system, it is used to classify handwritten digits (0–9).

Input Features

- Pixel values of handwritten digit images
- Image shape and patterns

Output

- Digit Class (0–9)

Sample Dataset:

Image Features	Digit
Pixel Pattern 1	0
Pixel Pattern 2	1
Pixel Pattern 3	5
Pixel Pattern 4	9

Working

1. Collect handwritten digit images.
2. Extract pixel features from each image.
3. Train the probabilistic generative model using the dataset.
4. The model learns the probability distribution of each digit class.
5. For a new handwritten image, the model calculates the probability for each digit and predicts the most likely class.

Example Prediction

Suppose a new handwritten image is given to the model.

The model predicts:

Digit = 7

Advantages

- Handles uncertainty effectively.
- Provides probability-based predictions.
- Works well for image classification tasks.
- Improves OCR system accuracy.

Conclusion

A Probabilistic Generative Model accurately classifies handwritten digits by learning the probability distribution of image features, making it useful in OCR applications such as digit recognition.

6. Customer Churn Prediction Using a Probabilistic Discriminative Model

Introduction

A Probabilistic Discriminative Model predicts the probability of a customer leaving a service based on input features. In the telecom industry, it is used to identify customers who are likely to stop using the service (customer churn).

Input Features

- Service Usage
- Call Duration
- Data Usage
- Monthly Bill

Output

- Churn
- No Churn

Sample Dataset

Service Usage	Data Usage (GB)	Monthly Bill (₹)	Churn
High	30	799	No
Low	5	299	Yes
Medium	15	499	No
Low	8	350	Yes

Working

1. Collect customer service usage data.
2. Train the probabilistic discriminative model using the dataset.
3. The model learns the relationship between customer behavior and churn.
4. It calculates the probability of a customer leaving the service.
5. Predict whether the customer will churn or not.

Example Prediction

Suppose a customer has:

- Service Usage = **Low**
- Data Usage = **6 GB**
- Monthly Bill = **₹320**

The model predicts:

Result = Churn

Advantages

- Predicts customer churn accurately.
- Helps telecom companies retain customers.
- Supports targeted marketing strategies.
- Improves customer satisfaction.

Conclusion

A Probabilistic Discriminative Model effectively predicts customer churn based on service usage data, enabling telecom companies to take preventive actions and improve customer retention.

7. Customer Purchase Prediction Using Bayesian Logistic Regression

Introduction

Bayesian Logistic Regression is a probabilistic classification algorithm that predicts the likelihood of an event while incorporating uncertainty. In online advertising, it is used to predict whether a customer will purchase a product after viewing advertisements.

Input Features

- Number of Advertisement Views
- Time Spent on Advertisement
- Customer Age
- Previous Purchase History

Output

- Purchase
- No Purchase

Sample Dataset

Ad Views	Time Spent (min)	Previous Purchase	Purchase
5	8	Yes	Yes
2	3	No	No
6	10	Yes	Yes
1	2	No	No

Working

1. Collect customer advertisement data.
2. Train the Bayesian Logistic Regression model using the dataset.
3. The model estimates the probability of a customer making a purchase.
4. It considers the relationship between advertisement features and purchase behavior.
5. Predict whether the customer will purchase the product.

Example Prediction

Suppose a customer has:

- Advertisement Views = **6**
- Time Spent = **9 minutes**
- Previous Purchase = **Yes**

The model predicts:

Result = Purchase

Advantages

- Provides probability-based predictions.
- Handles uncertainty effectively.
- Improves marketing decisions.
- Helps increase sales through targeted advertising.

Conclusion

Bayesian Logistic Regression predicts whether a customer will purchase a product after viewing online advertisements, helping businesses optimize advertising campaigns and improve sales.

8. Monthly Sales Forecasting Using Linear Regression

Introduction

Linear Regression is a supervised machine learning algorithm used to predict continuous values. In a retail company, it can forecast monthly sales based on advertising expenditure and seasonal demand.

Input Features

- Advertising Expenditure (₹)
- Seasonal Demand
- Previous Sales Data

Output

- Predicted Monthly Sales

Sample Dataset

Advertising (₹)	Seasonal Demand	Monthly Sales (₹)
50,000	High	5,00,000
30,000	Medium	3,50,000
70,000	High	6,50,000
20,000	Low	2,50,000

Working

1. Collect historical sales data.
2. Use advertising expenditure and seasonal demand as input features.
3. Train the Linear Regression model using the dataset.
4. The model learns the relationship between inputs and monthly sales.
5. Predict future monthly sales based on new input values.

Linear Regression Equation

$$\text{Sales} = b_0 + b_1(\text{Advertising}) + b_2(\text{Seasonal Demand})$$

Where:

- b_0 = Intercept
- b_1, b_2 = Model coefficients

Example Prediction

Suppose:

- Advertising Expenditure = **₹60,000**
- Seasonal Demand = **High**

The model predicts:

Monthly Sales = ₹5,80,000

Advantages

- Simple and easy to implement.
- Provides accurate sales forecasts.
- Helps in budgeting and inventory planning.
- Supports better business decisions.

Conclusion

Linear Regression effectively forecasts monthly sales by analyzing advertising expenditure and seasonal demand, helping retail companies improve planning and maximize profits.

9. News Article Classification Using Naïve Bayes

Introduction

Naïve Bayes is a supervised machine learning algorithm based on Bayes' Theorem. It is widely used for text classification. In a news portal, it classifies articles into categories such as **Sports, Politics, and Entertainment**.

Input Features

- Word frequencies
- Keywords in the article
- Article content

Output

- Sports
- Politics
- Entertainment

Sample Dataset

Keywords	Category
Match, Goal, Cricket	Sports
Election, Government, Policy	Politics
Movie, Actor, Music	Entertainment
Team, Tournament	Sports

Working

1. Collect labeled news articles.
2. Extract keywords and word frequency features.
3. Train the Naïve Bayes model using the dataset.
4. The model calculates the probability of each category.
5. Classify the article into the category with the highest probability.

Example Prediction

Suppose a news article contains the keywords:

- **Cricket**
- **Match**
- **Tournament**

The model predicts:

Category = Sports

Advantages

- Fast and efficient for text classification.
- Handles large text datasets well.
- Easy to implement.
- Provides accurate category predictions.

Conclusion

Naïve Bayes effectively classifies news articles into Sports, Politics, and Entertainment based on keywords and word frequencies, helping news portals organize content automatically.

10. Student Placement Prediction Using Bayesian Logistic Regression

Introduction

Bayesian Logistic Regression is a probabilistic classification algorithm used to predict binary outcomes while considering uncertainty. In a placement system, it predicts whether a student will be **placed** or **not placed** based on CGPA, coding skills, and aptitude scores.

Input Features

- CGPA
- Coding Skills Score
- Aptitude Test Score

Output

- Placed
- Not Placed

Sample Dataset

CGPA	Coding Skills	Aptitude Score	Placement
8.9	90	88	Placed
6.8	60	65	Not Placed
8.2	85	80	Placed
7.0	68	70	Not Placed

Working

1. Collect student academic and skill data.
2. Train the Bayesian Logistic Regression model using the dataset.
3. The model estimates the probability of a student getting placed.
4. It analyzes the relationship between CGPA, coding skills, and aptitude scores.
5. Predict whether the student will be placed or not.

Example Prediction

Suppose a student has:

- **CGPA = 8.5**
- **Coding Skills = 87**
- **Aptitude Score = 84**

The model predicts:

Result = Placed

Advantages

- Provides probability-based predictions.
- Handles uncertainty effectively.
- Supports placement decision-making.
- Improves prediction accuracy.

Conclusion

Bayesian Logistic Regression accurately predicts student placement outcomes using CGPA, coding skills, and aptitude scores, helping colleges and recruiters identify placement-ready candidates.